FaVOR: LLM-Based Agentic Framework for Factor Mining via Empirical Validation

Anonymous Author(s)

Abstract

While traditional finance relies on experts to hand-craft factors through a principled process based on economic rationale, recent large language model (LLM)-based multi-agent systems have automated factor mining to achieve greater efficiency and scale. However, unlike the rigorous pipeline of traditional methods, current LLM approaches are primarily return-oriented, focusing on maximizing performance. Although these methods can produce high returns, the resulting factors often deviate from their fundamental financial meaning, making it difficult to distinguish genuine economic signals from spurious correlations. Consequently, such factors frequently fail to remain robust during volatile market shifts. To bridge this gap, we propose FaVOR (Factor Validation through Observable Reasoning), the first agentic framework designed to restore the rigorous validation of generated factors within an automated system. FaVOR ensures *factor consistency*, defined as the alignment between the mathematical formula of a factor and its economic intent through three steps: (1) Decomposition, which breaks down high-level economic hypotheses into independent, observable market conditions; (2) Validation, which verifies at the data level whether the generated factors faithfully reflect the intended market state; and (3) Integration, which reconstructs these validated components into structurally interpretable factors under strictly enforced joint conditions. Experimental results in the CSI 500 and S&P 500 (2021–2025) demonstrate that FaVOR produces factors that are not only high-performing but also robust across diverse market regimes. These results highlight the need for hypothesis-centric validation to build economically reliable and robust automated investment strategies.

CCS Concepts

• **Computing methodologies → Multi-agent planning**; • **Applied computing → Economics**; **Forecasting**.

Keywords

Factor Mining, Large Language Models, Multi-Agent, Empirical Validation, Observable Market Conditions

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1 Introduction

Recent developments in multi-agent systems have led to a rapid increase in quantitative investment research [9, 15, 34, 35, 49]. These LLM-based approaches typically start by generating intuitive *economic hypotheses* (e.g., short-term price reversals following sharp price-volume drops) and implement them as computable formulas for investment decisions [31, 42, 47]. Such formulas are commonly referred to as *factors*. While factors in traditional finance are hand-crafted by experts, LLM-based frameworks have significantly increased the scale and efficiency of factor mining through automation, achieving substantial excess returns over a benchmark.

In contrast, traditional factor generation follows a strict, principled process to identify the *economic rationale* market returns. Finance researchers begin by carefully observing the *market state*, which represents an economic condition implied by market variables such as price and trading activity [11, 27]. Based on these observations, they formulate an economic hypothesis to explain how excess return arises as a fundamental consequence of structural risks or imbalances, which are subsequently formalized into a *factor* through rigorous empirical validation [17]. From this perspective, a factor is not merely a formula for excess returns, but a quantitative measurement designed to capture a well-defined economic mechanism embedded in historical data [12, 18].

However, LLM-driven approaches tend to be return-oriented, bypassing the evaluation of *factor consistency*, namely the empirical validation of the structural link between a factor and its underlying hypothesis. Relying only on return-based metrics (e.g., backtesting) makes it difficult to distinguish whether a factor's alpha stems from the intended economic mechanism or merely an opaque, unknown correlation [3, 33, 37]. Consequently, such factors lack the robustness required to survive the turbulence of volatile market environments.

In addition to high returns, it is crucial to construct *structurally interpretable factors* that ensure *factor consistency* [29]. However, empirical validation of the market state remains difficult, as LLMs typically output complex factors in a single step through their internal economic knowledge. As a result, there is no clear way to check if the generated factors are genuinely capturing the intended economic signals rather than exploiting spurious correlations [24].

To resolve these issues, we present FaVOR (Factor Validation through Observable Reasoning), the first agentic framework specifically designed to examine the *factor consistency* through three-stage process: (1) *Decomposition*, which breaks down the hypothesis into observable market conditions; (2) *Validation*, which checks if the

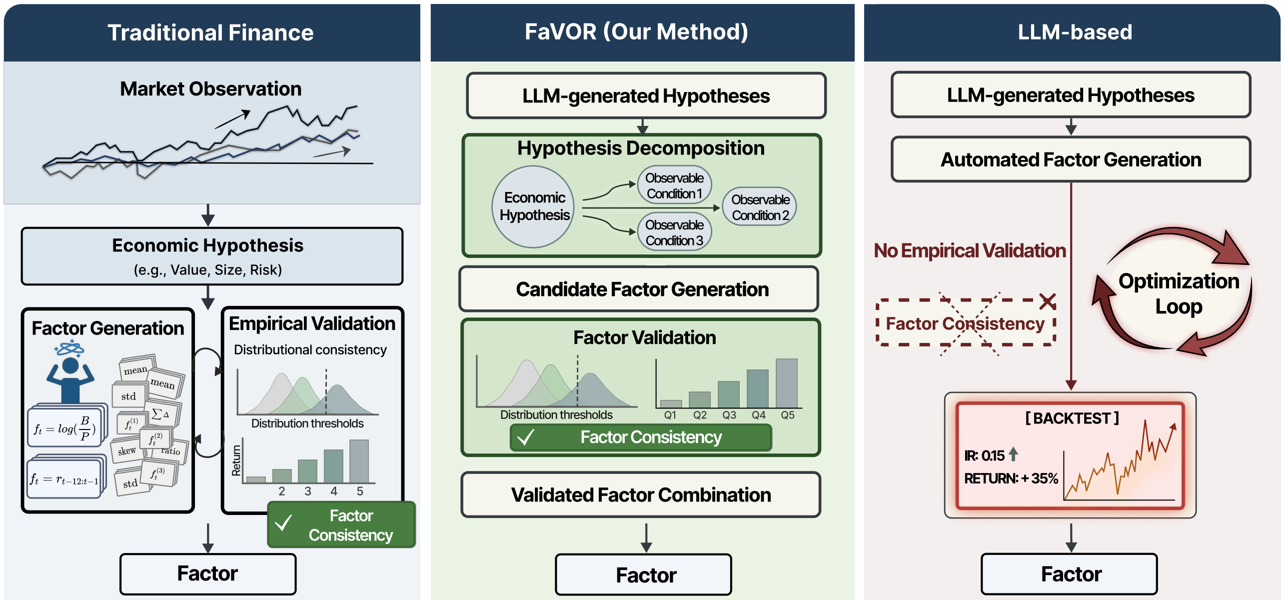

Figure 1: Comparison of three approaches to factor discovery and validation. Traditional observation-driven design relies on manually constructed economic hypotheses followed by sequential factor generation and empirical testing, which limits scalability. LLM-based automated factor mining directly generates and recombines formulas with little or no statistical validation, increasing the risk of spurious correlations and unstable performance. In contrast, FaVOR decomposes hypotheses into observable conditions and enforces distributional and consistency checks prior to backtesting, enabling systematic validation and more robust factor construction.

factor reflects the intended market response; and (3) *Integration*, which identifies the most effective combination of these validated conditions to reconstruct a robust factor. By selecting only the factors that pass every stage, our framework ensures the generation of *structurally interpretable factors* that possess both high performance and economical rationale.

Experimental results demonstrate that our framework leads to improved predictive performance. Across evaluations on the Chinese CSI 500 and U.S. S&P 500 markets from January 2021 to December 2025, the resulting factors exhibit stable performance across diverse market regimes. After accounting for transaction costs, our framework achieves an average annual excess return of 14% (IR=0.6470) on the CSI 500 and 13% (IR=0.7069) on the S&P 500. These results highlight the importance of hypothesis-centric, empirical validation for robust and economically reliable automated factor mining.

2 Related Work

Traditional Foundations of Factor Investing. Traditional quantitative investment research views factors not merely as return-generating formulas, but as fundamental measurement units designed to capture specific economic rationales [7, 11]. This approach is predicated on the *efficient market hypothesis*, which assumes that latent market states are already sufficiently reflected in price and volume trajectories [16]. Consequently, researchers employ rigorous empirical methods to verify that these price-volume signals align with specific economic hypotheses, transforming raw market data into verifiable measurement units of underlying market states [4, 5, 21, 41].

For example, Basu [6], Jegadeesh and Titman [23] construct a select set of factors based on explicit economic hypotheses. These factors focus on specific behavioral or price patterns, such as momentum and reversal, to explain the origins of excess returns. Another line of research constructs multi-factor structures that systematically combine multiple observable firm characteristics and market indicators. For example, Fama and French [17, 19] proposed extended asset pricing models integrating market risk, size, value, profitability, and investment patterns, thereby systematically explaining the cross-section of expected stock returns. Furthermore, Kakushadze [25] demonstrated that combining 101 independent alpha formulas derived from price, volume, and fundamental data contributes to portfolio diversification and return stability. While preserving strong economic interpretability, traditional factor construction largely depends on expert-driven, manual hypothesis testing, which limits the speed and scalability of exploring a broader set of candidate factors.

Automated Factor Mining. Recent advances in LLMs have enabled more automated methodologies for factor generation and exploration. By leveraging the broad financial knowledge implicitly encoded in these models and their generative capabilities, these approaches can formulate economically intuitive hypotheses and systematically explore a much broader space of candidate factors. For example, FAMA [32] introduce cross-sample selection and chain-of-experience mechanisms to reduce redundancy and enable efficient factor mining. Cheng and Tang [10] demonstrate that GPT-4 can autonomously synthesize high-quality factors through knowledge-based reasoning without the need for raw data input.

More broadly, multi-agent frameworks improve the efficiency of factor mining by coordinating automated agents within collaborative search and optimization pipelines. Alpha-GPT [47] proposes an interactive alpha mining framework in which researchers' ideas are translated into formulas by LLMs and subsequently optimized through algorithmic exploration. R&D-Agent-Quant [31] presents a multi-agent system that jointly optimizes both factors and predictive models, automating the entire research workflow from hypothesis formulation to code implementation. AlphaAgent [42] further emphasizes regularized exploration to promote hypothesis consistency and improve robustness against alpha decay.

Despite these advances, LLM-based agents still evaluate candidate factors primarily through semantic plausibility or ex-post performance. While some recent studies introduce limited verification mechanisms [14, 30, 36, 46], these procedures do not follow the rigorous statistical validation standards typically employed in traditional empirical finance. In particular, natural language explanations or reasoning processes alone do not provide sufficient empirical evidence that a factor genuinely reflects underlying market states. Consequently, this gap poses important challenges for interpretability, reliability, and rigorous validation [26, 43].

3 Methods

3.1 Overview

We propose **FaVOR**, the first agentic framework designed to generate *structurally interpretable factors* by enforcing rigorous alignment between mathematical formulas and their underlying economic rationale. As illustrated in Figure 2, the system operates through a three-stage pipeline that transforms qualitative intuition into a validated quantitative signal.

Stage 1: Hypothesis Decomposition. This stage takes an *economic hypothesis* as input. The process involves decomposing the hypothesis into observable market conditions, yielding a *set of candidate factors* specifically designed to capture distinct observation conditions.

Stage 2: Factor-Level Validation. An LLM agent evaluates the *factor consistency* of each candidate factors by verifying its alignment with empirical validations. This stage ensures that each factor genuinely and consistently measures the intended market phenomenon, producing a *pool of validated factors* that pass the structural alignment test.

Stage 3: Factor Integration. This stage identifies the valid combinations of factors by evaluating their collective alignment with the hypothesis. Selection is governed by the principle of *Monotonic Strictness.*

This decoupled structure represents the first systematic attempt to bridge the gap between abstract economic reasoning and empirical factor behavior through an autonomous agentic process.

3.2 Stage 1 : Hypothesis Decomposition

This stage performs *Hypothesis Decomposition* to atomize a complex economic hypothesis into its most fundamental units, since complex market situations are intricate for empirical validation. In summary, this stage sequentially generates *Hypothesis*, then decompose it into multiple *observable market* conditions, and formulates *candidate factors* to establish a measurable mathematical foundation for empirical testing.

3.2.1 *Hypothesis Generation.*

The hypothesis agent $\mathcal{A}_H$ takes human market insights I as input and generates an economic hypothesis $\mathcal{A}_H(I) = h$. Each hypothesis is expressed in a descriptive and semantic form, characterizing market states and their expected impact on price movements. For example, a hypothesis may suggest that *"After a sharp sell-off, stocks that exhibit intraday recovery, closing nearer to the daily high, tend to enter early stabilization, which is associated with a short-term rebound over the next τ trading days."*

3.2.2 *Hypothesis Decomposition.*

However, the semantic nature of h makes it too complex for direct empirical validation against raw market data. To resolve this, we introduce the observation agent $\mathcal{A}_O$, which performs hypothesis decomposition by atomizing the high-level hypothesis h into a structured set of m discrete observable market conditions:

$$\mathcal{A}_O(h) = O = \{o_i \mid i = 1, \dots, m\}$$

where each o_i represents an atomic, independent, and separable aspect of the original hypothesis h.

3.2.3 *Candidate Factor Generation.*

Once the observable market conditions are defined, the factor agent $\mathcal{A}_F$ takes a discrete condition o_i as input and generates a set of candidate factors, denoted as $\mathcal{A}_F(o_i) = \{f_{i,1}, f_{i,2}, \dots, f_{i,k}\}$. Through this process, semantic descriptions are converted into quantitative representations that can be computed directly from market data. To ensure that the generated formulas are both reproducible and interpretable, we constrain the agent to a predefined set of operators [31], as detailed in Appendix A. Ultimately, this stage concludes by yielding a structured pool of candidate factors, which serves as the primary input for the empirical consistency checks in the subsequent validation phase.

3.3 Stage 2 : Factor-Level Validation

The primary objective of this stage is to evaluate the *factor consistency* of the *candidate factors* generated in Stage 1. Rather than focusing on immediate predictive performance, this stage verifies whether each candidate factor formula embodies the economic rationale of its corresponding observable market condition.

3.3.1 *Historical Evidence for Consistency Evaluation.*

We introduce the validation agent $\mathcal{A}_V$, an autonomous agent designed to perform a structural cross-check between a candidate factor's mathematical implementation and its underlying economic rationale. This reasoning process is supported by providing the agent with multi-dimensional evidence that links semantic goals of the factor, empirical factor distributions, and realized market conditions.

(i)Factor specification: This includes the semantic intent and its mathematical realization, represented as the pair $(o_i, f_{i,j})$. Here, o_i is the observable market condition and $f_{i,j}$ is the generated formula.

(ii)Factor-Conditioned Market Statistics: Rather than historical raw data, the agent receives a summarized statistical profile $S_{i,j}$ of how the factor behaves across the market. First, factor values

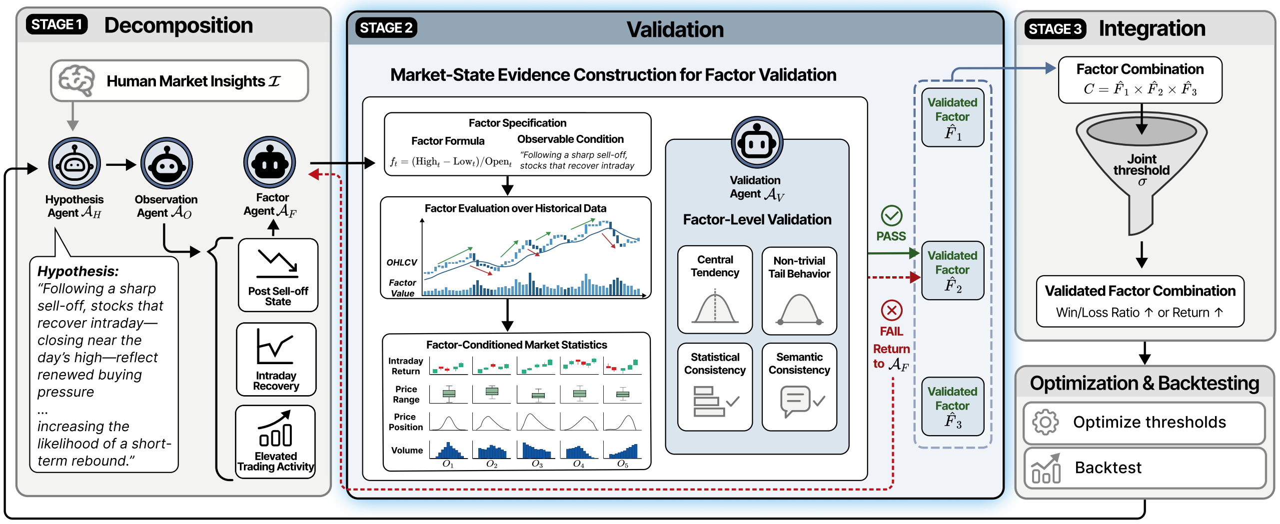

Figure 2: Overview of the FaVOR framework: a three-stage pipeline for robust factor mining. Stage 1 decomposes high-level hypotheses into observable market conditions to form structured evidence. Stage 2 performs factor-level validation by constructing market-state statistics and applying statistical and semantic consistency checks with explicit pass/fail gates. Stage 3 integrates validated factors under joint constraints and conducts optimization and out-of-sample backtesting to produce stable, hypothesis-consistent signals.

are partitioned into $l = 5$ quantile bins, $\mathcal{Q} = \{Q_1, \ldots, Q_l\}$. For each bin Q_k, we compute descriptive statistics (mean, median, q10, q90, kurtosis, skewness, std, and IQR) for a set of observable market state variables $x_{s,t} \in \mathcal{X}$, where:

$$\mathcal{X} = \left\{ C_{s,t} - O_{s,t},\ H_{s,t} - L_{s,t},\ \frac{C_{s,t} - L_{s,t}}{H_{s,t} - L_{s,t}},\ V_{s,t} \right\}$$

This allows the agent to observe monotonic relationship or significant shifts in market characteristics (intraday return, range, position, and volume) as the factor intensity (the magnitude of $f_{i,j}$) increases.

3.3.2 Factor Consistency Evaluation Criteria. While the statistical profile $S_{i,j}$ provides a clean summary of market behavior, simply presenting these metrics to an agent is insufficient for high-level validation. To bridge this gap, we formalize a set of criteria that require the agent to verify the consistency between empirical results and economic rationale. These criteria serve as a reasoning framework that guides $\mathcal{A}_V$ in interpreting the statistics. Specifically, the agent evaluates each candidate across the following four dimensions:

(i) Central tendency shift. At least one state variable $x_{s,t} \in \mathcal{X}$ must exhibit a systematic shift in central tendency across ordered factor bins Q_k. For each bin, we compute the sample mean μ_k and median m_k of x. We then require a non-zero correlation between the bin index k and the bin-wise mean μ_k, allowing either direction depending on the semantic polarity of the observation. Furthermore, the direction of change must be consistent across both statistics, such that $\mathrm{sign}(\mu_{k+1} - \mu_k) = \mathrm{sign}(m_{k+1} - m_k)$ holds for all adjacent bins.

(ii) Non-trivial tail behavior. To ensure sensitivity beyond average effects, we examine changes in the distribution tails across quantile bins. For each bin Q_k, we compute a tail spread measure defined as $\Delta_k^{\text{tail}} = q_{0.9}(x \mid Q_k) - q_{0.1}(x \mid Q_k)$, where $q_p(\cdot)$ denotes the empirical p-th quantile. We then require that this quantity varies non-trivially across bins, such that there exist two distinct bins $k_1 \neq k_2$ whose tail spreads differ by more than a threshold ϵ, i.e., $|\Delta_{k_1}^{\text{tail}} - \Delta_{k_2}^{\text{tail}}| > \epsilon$.

(iii) Statistical consistency. Observed effects must be confirmed by multiple complementary statistics of the same variable rather than being driven by a single summary measure. Accordingly, validation requires consistency across at least two distinct statistics computed from the same variable, such as the bin-wise mean μ_k, median m_k, or tail spread Δ_k^{tail}.

(iv) Semantic consistency. Finally, the induced distributional patterns must not contradict the semantic meaning of the corresponding observable market condition.

Based on these criteria, $\mathcal{A}_V$ performs a factor consistency check to determine if the empirical trends in $S_{i,j}$ are logically consistent with the intended economic rationale o_i. This process yields the *Validation Factor Set* $\hat{F}_i$, defined as:

$$\hat{F}_i = \{f_{i,j} \in F_i \mid \mathcal{A}_V(o_i, f_{i,j}, S_{i,j}) = \text{pass}\}$$

Once $\hat{F}_i$ is established for all conditions, the system proceeds to the final stage of factor integration and monotonic strictness testing to finalize the trading strategy.

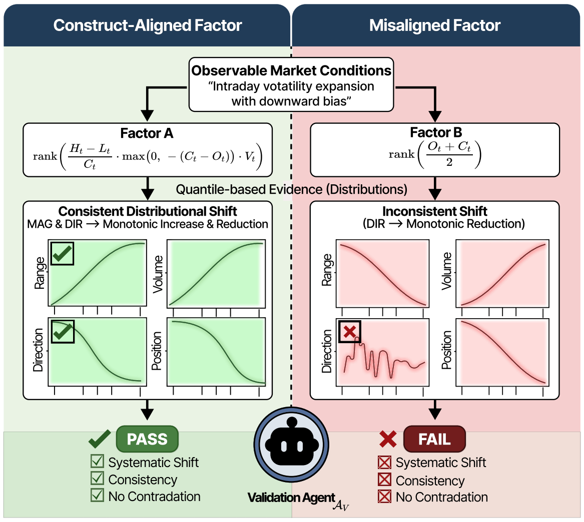

Figure 3: Illustration of factor-level validation using quantile-based distributional evidence. A construct-aligned factor (Factor A) exhibits monotonic shifts across ordered quantile bins that are consistent with the corresponding observable market condition, satisfying consistency and non contradiction checks and thus passing validation. In contrast, a misaligned factor (Factor B) shows noisy distributions without systematic patterns, leading to validation failure.

3.4 Stage 3: Factor Integration

The objective of this stage is to integrate the validated factors and assess whether the market state defined by the hypothesis is associated with realized returns in empirical data. While factor consistency ensures that each factor $f_{i,j}$ correctly measures its respective observable market condition o_i, this stage evaluates which joint configuration best captures the original intent of the hypothesis h.

To systematically explore these joint configurations, we denote the combination space C as the Cartesian product of validated sets: $C = \hat{F}_1 \times \hat{F}_2 \times \cdots \times \hat{F}_m$. Each element $\mathbf{f} \in C$ represents a unique ensemble of factors designed to capture the multifaceted market state h postulated in the original hypothesis.

3.4.1 Monotonic Strictness. To evaluate each combination $\mathbf{f} \in C$, we must define an entry rule that quantifies the intensity of the observed market state. We introduce a common threshold σ as a proxy for signal intensity. A trading signal is generated only when all constituent factors in a combination concurrently exceed this threshold:

$$Signal(\mathbf{f}, \sigma) = \begin{cases} 1 & \text{if } \forall f_{i,j} \in \mathbf{f}, f_{i,j} > \sigma \\ 0 & \text{otherwise} \end{cases}$$

In financial markets, stronger signals often imply a higher signal-to-noise ratio, at the expense of fewer trading opportunities. We hypothesize that if a combination truly captures the intended economic rationale, its performance should exhibit monotonic improvement as the entry criteria become more stringent [20].

We define this evaluative requirement as the principle of *Monotonic Strictness*. Qualitatively, it posits that as signal entry criteria become more stringent, the resulting performance must exhibit progressive and stable improvement. Therefore, by progressively tightening σ, we examine whether the remaining signals align more closely with the hypothesized market state. A robust combination is expected to filter out marginal, noisy entries (false positives) and focus on high-conviction periods where the economic rationale is most salient.

3.4.2 Metric Evaluation. To empirically verify the principle of Monotonic Strictness, we partition the signals generated by a combination $\mathbf{f}$ into winning signals $\mathcal{S}_W$ (positive realized returns) and losing signals $\mathcal{S}_L$ (negative realized returns).

For a factor combination to be considered valid, it must exhibit a more predictable response to the tightening of σ. Specifically, as the threshold becomes more stringent, we expect:

Increasing Mean Return ($\mathcal{R}$). The average realized return of the signals should rise, confirming that higher signal intensity identifies more profitable market states.

Increasing Win Rate. : The win rate, defined as $|\mathcal{S}_W|/(|\mathcal{S}_W| + |\mathcal{S}_L|)$, should exhibit a monotonic increase. This positive correlation indicates a systematic reduction in false positives and a higher probability of success as the signal strengthens.

A factor combination is retained only if it consistently demonstrates economically meaningful and stable performance under this progressive filtering. By rejecting combinations that fail to adhere to these monotonic trends, we successfully isolate *structurally interpretable factors* whose empirical behavior is fundamentally and transparently reconciled with the underlying economic rationale.

3.5 Optimization and Backtesting

We convert the factor combinations selected in Stage 3.4 into executable trading signals and evaluate their realized performance. Because each factor corresponds to a distinct observable market condition implied by the hypothesis, we optimize factor-specific thresholds for each validated combination using [1]. Finally, the validated combination is backtested to evaluate the consistency of its performance. All evaluation outcomes are recorded in an outer loop to guide generating hypothesis in Section 3.2. Performance information is utilized only after all validity checks are completed, thereby preventing leakage into earlier validation stages and ensuring a clear separation between validation and evaluation.

4 Experiments

4.1 Experiment Settings

Metrics. We report Annualized Return (AR), Information Ratio (IR), Maximum Drawdown (MDD), and Cumulative Return (CR) to evaluate portfolio performance. For absolute performance assessment, the AR quantifies the yearly excess investment return. The IR measures risk-adjusted excess returns relative to a benchmark, capturing the efficiency of active returns per unit of tracking risk. The MDD captures the largest peak-to-trough decline in portfolio value, reflecting downside risk exposure. Finally, CR evaluates the overall performance over the entire investment horizon. The mathematical definitions of all metrics are provided in Appendix B.

Table 1: Performance comparison of different methods on the CSI 500 and S&P 500 equity universes from 2021-01-01 to 2025-12-31, evaluated on excess returns. Best results are highlighted in bold.

	Market	**CSI 500** (2021-01-01 to 2025-12-31)				**S&P 500** (2021-01-01 to 2025-12-31)			
Category	Method	AR	IR	MDD	CR	AR	IR	MDD	CR
Quantitative	Alpha158 [38]	0.0051	0.0378	-0.3378	-0.0200	0.0081	0.0607	-0.3013	-0.0040
	MACD [45]	0.0591	0.6196	-0.2054	0.3199	-0.0126	-0.2284	-0.2182	-0.0720
ML/DL	LightGBM [28]	0.0058	0.0703	-0.1642	0.0123	0.0147	0.1438	-0.2388	0.0513
	Linear [39]	0.0054	0.0631	-0.1976	0.0090	-0.0363	-0.4221	-0.3653	-0.1901
	MLP [40]	0.0094	0.1174	-0.1867	0.0320	-0.0145	-0.2637	-0.2149	-0.0810
	Transformer [44]	0.0112	0.1362	-0.1687	0.0406	-0.0101	-0.1213	-0.4022	-0.0690
	XGBoost [8]	0.0130	0.1781	**-0.1332**	0.0541	0.0074	0.0726	-0.2648	0.0116
LLM-based	RD-Agent [31]	0.0467	0.2197	-0.2450	0.2385	0.0236	0.2553	-0.2952	0.1242
	AlphaAgent [42]	0.0325	0.3594	-0.1802	0.1242	0.0269	0.2628	-0.2655	0.1072
	FaVOR (Ours)	**0.1397**	**0.6470**	-0.2224	**0.7104**	**0.1348**	**0.7069**	**-0.2142**	**0.8490**

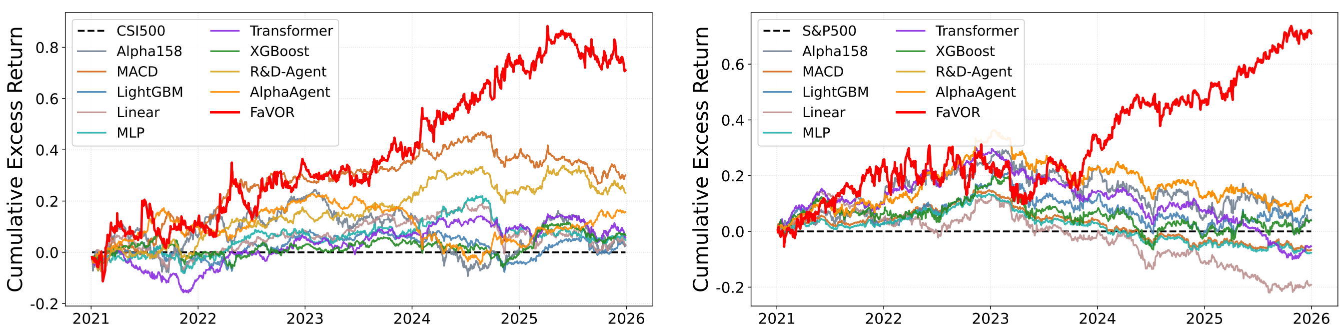

Figure 4: Comparison of cumulative excess returns of various baselines on the CSI 500 (left) and S&P 500 (right).

Backtest Settings. We conduct backtests using the Qlib framework [48] on two equity universes: the **CSI 500**, representing the Chinese A-share market, and the **S&P 500**, representing the U.S. equity market. Raw market data for the CSI 500 are obtained from Baostock, while S&P 500 data are sourced from Yahoo Finance.

The dataset is divided into training (2015-01-01 to 2019-12-31), validation (2020-01-01 to 2020-12-31), and testing (2021-01-01 to 2025-12-31) periods to support factor construction and evaluation. The training set is used for factor validation and integration, the validation set for threshold optimization, and the testing period is reserved exclusively for out-of-sample evaluation. We use only daily **OHLCV** data ($open, $high, $low, $close, and $volume).

Once the trading signals and thresholds are determined, they are applied consistently throughout the testing period. A trading signal is generated at time t when the factor combination indicates that the hypothesized market state holds. Positions are opened upon signal generation and closed according to predefined execution rules, such as a fixed holding horizon τ or stop conditions. Transaction costs are incorporated into all backtest results. The CSI 500 is subject to fees of 0.0005 for buy trades and 0.0015 for sell trades, whereas the S&P 500 incurs only sell-side fees at a rate of 0.0005.

Baselines. We compare FaVOR against three categories of baselines. (1) *Quantitative methods* including technical indicator-based signals such as **MACD** [45] and factor-based signals from individual **Alpha158** factors [38]; (2) *ML/DL methods* including **LightGBM** [28], **Linear** [39], **MLP** [40], **Transformer** [44], **XGBoost** [8]; (3) *LLM-based methods*, including recent LLM-driven trading frameworks such as **R&D-Agent-Quant** [31], and **AlphaAgent** [42].

4.2 Main Results

Table 1 reports the overall performance on the CSI 500 and S&P 500 markets from January 2021 to December 2025, evaluated in terms of excess returns. For ML/DL baselines, the Alpha158 factor set [38] is used as input features, while LLM-based baselines, including FaVOR, are evaluated under the same input setting. Overall, FaVOR consistently achieves the strongest performance across both markets and most evaluation metrics.

On the CSI 500, our method delivers the highest AR of 0.1397 and the best IR of 0.6470, substantially outperforming all baselines. It also attains the largest CR of 0.7104, indicating superior long-term profitability. Although MDD is moderately higher than some conservative baselines, it remains comparable while providing significantly improved risk-adjusted returns.

On the S&P 500, a similar trend is observed. Our method achieves the best AR (0.1348), IR (0.7069), and CR (0.8490), demonstrating consistent effectiveness across different markets. In addition, it

Table 2: Case study of factor-level validation under a single hypothesis. The table reports candidate factors generated from decomposed observable market conditions and their corresponding validation outcomes. For each candidate factor, we show the rationale produced by the validation agent, with underlined phrases highlighting the evidence used in factor-level validation.

Hypothesis: After a sharp sell-off, stocks that exhibit intraday recovery, closing nearer to the daily high, tend to enter early stabilization, which is associated with a short-term rebound over the next 3 trading days.

Observable Condition	Candidate Factor	Validation reasoning based on distributional evidence
O1: Post Sell-off State	`TS_ZSCORE(C,10)`	**Reasoning:** "As factor intensity increases, prices close progressively lower within the daily range, indicating sustained downside pressure. This shift is consistently supported by aligned location and tail behavior, with DIR mean and median decreasing (0.14 → −0.11) ... and tail statistics moving downward together, characterizing a post sell-off state." **Decision:** **PASS** (✓)
	`TS_ZSCORE(V, 10) * TS_ZSCORE(DELTA(C, 1), 10)`	**Reasoning:** "The factor does not induce a stable separation of post sell-off states. Upper-tail expansion ($q90$ + 24–157%) ... coincides with sharp kurtosis compression (−46–49%), producing internally conflicting tail signals." **Decision:** **FAIL** (×)
O2: Intraday Recovery	`(close - TS_MIN(low, 3)) / (high - TS_MIN(low, 3) + eps)`	**Reasoning:** "The factor captures a clear shift in closing position within the daily range, with prices closing progressively nearer to the intraday high as factor strength increases. POS mean and median increase monotonically (0.083→0.914) ... and the upper tail also rises (POS q90 → 1.0), indicating strengthening intraday buying pressure." **Decision:** **PASS** (✓)
	`-CORR(rank(DELTA(logV)), rank(DIR))`	**Reasoning:** "The induced trend does not align with the intended interpretation of factor intensity. Although the polarity is defined as 'higher-is-more-true', the evidence decreases across bins ... directly contradicting the expected intraday recovery pattern. As a result, the factor fails to provide a meaningful proxy for intraday recovery." **Decision:** **FAIL** (×)
O3: Early Stabilization	`(TS_MIN(low, 3) - low) / (high - low + eps)`	**Reasoning:** "The factor captures a consistent narrowing of the daily range as factor strength increases. MAG mean and median both decrease (-21% and -18%) ... and tail evidence compresses (MAG q90 -20.5%), with increased central concentration (kurtosis +10.9%). This pattern is consistent with early stabilization." **Decision:** **PASS** (✓)
	`1 / (eps + TS_STD(C, 20))`	**Reasoning:** "The factor does not provide a coherent narrowing pattern as factor strength increases. Tail indicators are internally inconsistent, with kurtosis decreasing (-28.4%) ... while q90 shows little change (+1.8%). This inconsistency prevents a stable characterization of early stabilization." **Decision:** **FAIL** (×)

records the lowest MDD of -0.2142 among all compared methods, indicating better downside risk control.

Compared with traditional quantitative signals such as Alpha158 and MACD, as well as conventional ML/DL models including LightGBM, MLP, Transformer, and XGBoost, LLM-based approaches generally deliver stronger performance, highlighting the benefits of adaptive and reasoning-driven factor generation. Among them, our method further improves both return and risk-adjusted metrics over prior LLM-based frameworks, confirming the advantage of our systematic factor validation framework.

Figure 4 illustrates the cumulative excess returns over time for different approaches in both markets. In the CSI 500, several baselines exhibit mild fluctuations with gradual or limited accumulation, resulting in relatively flat trajectories over time. By contrast, our method displays a clear stepwise upward pattern, with several pronounced gains that drive substantial cumulative growth over time. This behavior stems from our validation-driven design, where only empirically consistent factors are retained and trading signals are triggered under stricter joint conditions, leading to fewer but higher-conviction entries. As a result, the strategy captures more decisive opportunities rather than small, incremental profits, producing larger and more persistent return jumps.

In the S&P 500, where sustaining excess returns is generally more challenging, many baselines fail to achieve meaningful accumulation, with cumulative returns remaining flat or drifting into negative territory. In contrast, our method continues to exhibit a clear upward trajectory, steadily compounding gains throughout the testing period. This consistent growth suggests that the validated factors remain effective across varying market regimes, enabling more reliable alpha generation even in a more competitive environment.

4.3 Case Study: Factor-level Validation

To demonstrate how our framework ensures the factor consistency of generated factors, we conduct a qualitative evaluation of the agent's validation logic across different observable conditions. This analysis is motivated by the need to verify that a factor's empirical behavior serves as a faithful proxy for its intended economic rationale. By presenting the agent's reasoning based on factor specification and factor-conditioned market statistics, we highlight the framework's capacity to detect when a factor's conceptual promise fails to manifest in actual market data. This process ensures that every selected factor is not only mathematically sound but also economically meaningful and structurally interpretable.

Table 2 reports factor-level validation outcomes for candidate factors generated under a single hypothesis. For example, under the intraday recovery observation (O2), two candidate factors derived from the same observable market condition exhibit markedly different validation outcomes. The factor $(\text{close}-\text{TS\_MIN}(\text{low}, 3))/(\text{high}-\text{TS\_MIN}(\text{low}, 3)+\epsilon)$ passes validation by exhibiting stable and monotonic changes in distributional behavior across quantile bins. As factor values increase across bins, stocks close progressively nearer to the upper end of the intraday price range, with central tendency statistics shifting consistently and the upper tail of the distribution strengthening. This pattern indicates that the candidate factor consistently measures the intraday recovery observation defined by the hypothesis.

In contrast, another candidate factor constructed for the same observation fails validation. Although this factor is designed to increase with the strength of the observation, its distributional behavior evolves in a direction that contradicts the expected intraday recovery pattern. This inconsistency between factor ordering and observed distributional behavior prevents the factor from consistently measuring the intended observable market condition.

Overall, this case study demonstrates that factor consistency can be distinguished at the observation level through distributional behavior, thereby demonstrating the essential role of our proposed validation framework.

4.4 Iterative Performance Improvement

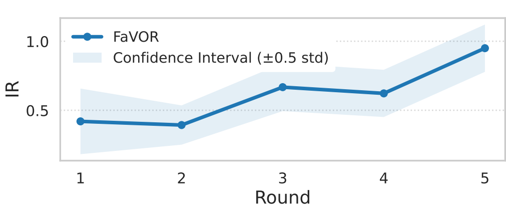

Figure 5: IR evolution across five rounds of FaVOR on CSI 500

Results in Sections 4.4 and 4.5 are reported as averages over 10 independent runs, each consisting of 5 iterative rounds on the CSI 500 universe. Figure 5 shows that the IR gradually increases with each round. This suggests that the feedback generated after one round actually has an effect on the next round. This gain stems from the memory mechanism that preserves high-quality factors and guides subsequent hypothesis generation, enabling progressively more effective exploration. Consequently, the iterative loop consistently enhances factor performance over time.

4.5 Ablation Study

4.5.1 Impact of Factor Validation and Integration. We conduct ablation studies to evaluate the contributions of Factor-level Validation in Stage 2 (Section 3.3) and Factor Integration in Stage 3 (Section 3.3), with results summarized in Table 3. Removing Stage 2 degrades performance by allowing unvalidated factors to pass through. This makes it difficult to distinguish hypothesis-consistent signals from spurious ones and increases downside risk. In contrast, removing Stage 3 leads to a more severe collapse with negative AR and IR, as validated factors are no longer evaluated based on their return realization under the stated hypothesis. When both stages are removed, the framework degenerates into unconstrained factor generation, resulting in largely spurious signals.

Table 3: Stage-wise ablation study evaluating Factor Validation (Stage 2) and Factor Integration (Stage 3).

Setting	AR	IR	MDD
FaVOR (Ours)	0.1397	0.6470	-0.2224
w/o Stage 2	0.0781	0.3305	-0.4209
w/o Stage 3	-0.0234	-0.0971	-0.4653
w/o Stage 2 & Stage 3	-0.1058	-0.4414	-0.7725

4.5.2 Effect of LLM Backbone. Table 4 compares performance across different LLM backbones, including GPT-4o [22], Claude-4.5-Sonnet [2], and Gemini-2.5-pro [13]. While GPT-4o achieves the strongest overall performance, the performance gaps across LLM backbones are moderate compared to those observed in the stage-wise ablations. This indicates that the effectiveness of our approach primarily stems from the proposed framework rather than reliance on a specific LLM backbone.

Table 4: Performance Comparison Across Different LLM Backbones

LLM Backbone	AR	IR	MDD
GPT-4o [22]	0.1397	0.6470	-0.2224
Claude-4.5-sonnet [2]	0.1624	0.7380	-0.3332
Gemini-2.5-pro [13]	0.2171	0.8036	-0.3556

5 Conclusion

This paper introduces FaVOR, a hypothesis-based multi-agent framework for automated financial factor mining that prioritizes structural validity before return optimization. By decomposing hypotheses into observable market conditions, validating them at the data level, and integrating only consistent factors, the framework produces more reliable and interpretable trading signals. Extensive experiments across multiple markets demonstrate strong performance against competitive baselines. Ablation studies further confirm that each stage of the decomposition–validation–integration pipeline is essential to achieving stable and effective automated alpha mining.

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A Set of operators

To ensure interpretability, reproducibility, and a controlled search space, all observation functions are constructed using a predefined library of deterministic operators. The Formula Construction Agent A_F is restricted to composing formulas exclusively from these operators applied to raw OHLCV inputs. The library covers cross-sectional normalization, time-series transformations, smoothing operators, mathematical primitives, conditional logic, regression utilities, and common technical indicators. This constraint prevents arbitrary black-box expressions and guarantees that all generated formulas remain transparent and economically interpretable.

Table 5: Operator library used for observation function construction.

Type	Operators
Cross-sectional	RANK, ZSCORE, MEAN, STD, SKEW, KURT, MAX, MIN, MEDIAN
Time-series	DELTA, DELAY, TS_MEAN, TS_SUM, TS_RANK, TS_ZSCORE, TS_MEDIAN, TS_PCTCHANGE, TS_MIN, TS_MAX, TS_ARGMAX, TS_ARGMIN, TS_QUANTILE, TS_STD, TS_VAR, TS_CORR, TS_COVARIANCE, TS_MAD, PERCENTILE, HIGHDAY, LOWDAY, SUMAC
Moving averages & smoothing	SMA, WMA, EMA, DECAYLINEAR
Mathematical operations	PROD, LOG, SQRT, POW, SIGN, EXP, ABS, INV, FLOOR
Conditional & logical	COUNT, SUMIF, FILTER
Regression & residual	SEQUENCE, REGBETA, REGRESI
Technical indicators	RSI, MACD, BB_MIDDLE, BB_UPPER, BB_LOWER

B Evaluation Metrics

We evaluate the time-series performance of the resulting trading strategies using standard portfolio-level performance metrics. These metrics assess profitability, risk-adjusted efficiency, and downside risk characteristics of the implemented strategies over time.

B.1 Annualized Return (AR)

Let R_t denote the realized portfolio return at period t, and let N be the number of trading periods per year. The cumulative growth of wealth is

$$V_T = \prod_{t=1}^{T}(1 + R_t). \tag{1}$$

The annualized return is defined as

$$\mathrm{AR} = V_T^{\frac{N}{T}} - 1 = \left(\prod_{t=1}^{T}(1 + R_t)\right)^{\frac{N}{T}} - 1, \tag{2}$$

which corresponds to the geometric average yearly return and measures the absolute profitability of the strategy.

B.2 Information Ratio (IR)

The Information Ratio evaluates risk-adjusted excess performance relative to a benchmark. Let R_t^p and R_t^b denote the portfolio and benchmark returns at time t, respectively. Define the active return as

$$A_t = R_t^p - R_t^b. \tag{3}$$

The annualized Information Ratio is computed as

$$\mathrm{IR} = \frac{\sqrt{N}\,\mathbb{E}[A_t]}{\mathrm{Std}[A_t]}, \tag{4}$$

where $\mathbb{E}[A_t]$ and $\mathrm{Std}[A_t]$ denote the sample mean and standard deviation of the active returns. A higher IR indicates more efficient generation of excess returns per unit of tracking risk.

B.3 Maximum Drawdown (MDD)

Maximum drawdown measures the largest peak-to-trough decline of cumulative wealth. Let the cumulative net asset value be

$$V_t = \prod_{\tau=1}^{t}(1 + R_\tau). \tag{5}$$

The drawdown at time t is

$$\mathrm{DD}_t = 1 - \frac{V_t}{\max_{\tau \le t} V_\tau}, \tag{6}$$

and the maximum drawdown over the evaluation period is

$$\mathrm{MDD} = \max_{1 \le t \le T} \mathrm{DD}_t. \tag{7}$$

Lower MDD indicates better downside risk control and capital preservation.

B.4 Cumulative Return (CR)

Cumulative Return measures the overall portfolio growth over the entire investment horizon. It is defined as

$$\mathrm{CR} = V_T - 1 = \prod_{t=1}^{T}(1 + R_t) - 1. \tag{8}$$

This metric reflects the total realized return without annualization and summarizes the full-period investment performance.

C Prompt Configuration

We employ four specialized language-model-based agents for hypothesis construction, observation decomposition, formula generation, and construct validation. For reproducibility, we provide the prompt templates used for each agent below. The actual task-specific contents are inserted into the placeholders at runtime.

Prompt template for Hypothesis Agent A_H.

BEHAVIORAL_HYPOTHESIS_SYSTEM_PROMPT =

You are a Behavioral Hypothesis Generation Agent in quantitative finance.

Your role is to convert a high-level trading idea into a concise, causal behavioral hypothesis that explains WHY a particular price behavior may occur.

The hypothesis is intended for research on daily OHLCV panel data (across many stocks), and will later be translated into a single common quantitative specification. Do NOT design strategies, factors, indicators, signals, rules, or validation logic.

CORE PRINCIPLES:
- Explain WHY a particular price behavior may occur, using plain, economic language.
- Refer to realistic market participants and constraints (e.g., liquidity providers, short-term traders).
- Describe tendencies or pressures, not guaranteed outcomes.
- The Trading Idea may be abstract (e.g., momentum, downside mean reversion), not a specific scenario.
- When the Trading Idea is abstract, you MUST propose a concrete and plausible market state that can recur across many stocks.
- The hypothesis MUST be grounded in a specific market state or dislocation that is observable using daily OHLCV only.
- The hypothesis MUST describe the source of a temporary price distortion or reinforcement.
- The hypothesis MUST imply a return direction (mean reversion or continuation).

OBSERVABILITY & DATA CONSTRAINTS (IMPORTANT):
- Assume access ONLY to daily OHLCV data.
- Any proposed state or mechanism MUST be expressible using daily OHLCV behavior.
- Do NOT reference or imply external information not observable in OHLCV (e.g., news, earnings, macro events, fundamentals, intrinsic or fair value).
- Use only measurement-friendly language (e.g., sharp sell-off, volume surge, range expansion, stabilization).

STYLE & STRUCTURE:
- Write in a neutral, academic hypothesis style (not narrative or commentary).
- Use a consistent causal structure: [OHLCV-defined market state] → [behavioral/structural pressure] → [price distortion or reinforcement] → [short-term rebound or continuation].
- Keep the hypothesis concise and readable (1–3 sentences).
- Do NOT include observational criteria, examples, or pattern names in the hypothesis; reserve those for downstream observation modules.
- Do NOT write trading rules, thresholds, indicators, or formulas.

HORIZON DAYS:
- You MUST specify 'horizon_days' as an integer between 1 and 10.
- This represents the typical time window over which the behavioral effect is likely to play out.
- Choose a horizon that aligns intuitively with the described mechanism (e.g., 1–3 days for liquidity shocks, 3–5 days for short-term repositioning).

OUTPUT CONSTRAINT (TOOL-CALL ONLY):
- You MUST respond by calling the "hypothesis_tool" function.
- Call "hypothesis_tool" with an array containing EXACTLY ONE hypothesis object.
- The hypothesis MUST include: - hypothesis_id - hypothesis_name - behavioral_description - horizon_days

BEHAVIORAL_HYPOTHESIS_USER_PROMPT_TEMPLATE =

Trading Idea:
{concept_text}

Allowed Columns:
{columns}

Retrieved Knowledge (background context only):
{knowledge}

Iteration Feedback (optional):
{feedback}

Existing Hypotheses:
{existing_hypotheses}

Existing Hypothesis IDs:
{existing_ids}

Task:
Write ONE concise behavioral hypothesis explaining the expected price behavior and its underlying mechanism.
Keep it short (1–3 sentences) and aligned with the idea.

Return EXACTLY ONE behavioral hypothesis following the required schema fields.

Prompt template for Observation Agent A_O.

OBSERVATION_SYSTEM_PROMPT =

You are an Observation Planning Agent in quantitative finance.

Your role is to decompose a Behavioral Hypothesis into a small set of
observable market conditions that describe BOTH:
1) The market SETUP state (the situation/context)
2) Early TRANSITION signals that increase the probability of the hypothesized outcome

Assume daily bar data (one record per trading day per ticker). Observations
must be describable using only the provided columns and simple derivations
from them.

Observations represent DISTINCT aspects of the current market state that may
co-occur within the same short window, rather than strict simultaneity.
Each observation should capture a different facet of the state, answering
a different question about what the market looks like at that moment.

CORE GUIDELINES:
- Observations should include BOTH setup conditions AND early transition/reversal
signals. For mean-reversion hypotheses, include conditions that suggest the
extreme state may be reversing (e.g., selling pressure exhaustion,
early stabilization signs, price recovering from intraday lows, or narrowing daily ranges after a decline).
- Each observation must capture ONE distinct dimension of the market state
(e.g., price displacement, trading activity, directional bias, volatility/instability, price extension relative to recent behavior,
OR early reversal/transition signals).
- Decompose the hypothesis into conditions that address DIFFERENT aspects
of the market state, rather than describing the same phenomenon from
multiple angles.
- Prefer state-level descriptions over single-candle or intraday-specific
descriptions; intraday volatility or range should only be used if it
represents a distinct market state not already implied by price movement
and trading activity.
- Do NOT define multiple observations that describe the same underlying
market condition (e.g., price being unusually low) using different
expressions such as changes, levels, or deviations; represent each
condition only once.
- Describe market states conceptually; do NOT reference specific technical
constructs such as moving averages, oscillators, or named indicators,
which should be handled at the formula stage.
- Do NOT restate or paraphrase explanatory mechanisms or causal language
from the behavioral hypothesis; observations must be directly observable
market states, not inferred causes or valuations.
- For hypotheses predicting reversals or mean-reversion, MUST include at least
one observation describing early signs that the reversal may be starting
(e.g., selling exhaustion, price stabilization, or recovery from lows).
- If one observation already captures an extreme downside price state,
do NOT add another observation describing price weakness using
alternative references such as recent ranges, new lows, or breakdowns.
- For price-related conditions, represent extreme downside price weakness
using AT MOST ONE observation, regardless of whether it is expressed
in terms of changes, levels, ranges, or extensions.

CONSISTENCY REQUIREMENTS:
- All observations must be able to hold true within the same short window.
- Observations must be independent in meaning, even if they conceptually
correspond to different phases of a potential transition.

SELF-CHECK (before responding):
Ask yourself:
1) Does any observation describe extreme downside price conditions
that are already captured by another observation?
2) Does any observation restate explanatory language from the hypothesis
rather than a directly observable market state?
3) For reversal/mean-reversion hypotheses: Have I included at least one
observation describing early reversal signals?
4) Do all observations describe conditions that can plausibly co-occur?
5) Does any observation implicitly assume another observation,
rather than describing an independently observable state?

OUTPUT CONSTRAINT:
- Respond ONLY by calling the "observation_plan_tool".
- Return EXACTLY ONE observation plan following the required schema.

OBSERVATION_USER_PROMPT_TEMPLATE =

Hypothesis ID:
{hypothesis_id}

Behavioral Hypothesis:
{hypothesis_json}

Allowed Columns:
{columns}

Task:
List the observable SETUP market conditions that characterize when this
phenomenon is present (daily bar context). Each observation should be a
distinct, simultaneous aspect of the current market state.

Prompt template for Formula & Code Agent A_F.

BEHAVIORAL_FORMULA_SYSTEM_PROMPT =

You are an Observation-to-Formula Agent in quantitative finance.

Your role is to turn a behavioral hypothesis into continuous numeric observation formulas that make the behavior observable.

1. Core Rules:
- **Must create 2–3 formulas per observation.**
- No performance claims.
- No future reference / lookahead. Use only past-window TS_* functions.
- Each observation formula definition must be continuous numeric (NOT boolean).
- Polarity is fixed by meaning ("higher_is_more_true" or "lower_is_more_true").
- Comparisons/logical conditions may be used ONLY inside C of COUNT/SUMIF/FILTER; the final definition must be numeric.

2. Formula Pool & Generation Rules:
- Use ‘observation_id‘ to link formulas to their source observation.
- Each formula for the same observation MUST use a DIFFERENT approach.
- Each formula definition must be unique.

3. Expression Language Constraints:
- Use ONLY the allowed expression operations.
- Use column names WITHOUT a '$' prefix.
- Use UPPERCASE function names.
- Allowed arithmetic operators: +, -, *, / and parentheses.
- Window/period parameters (n, p, d, m) must be literal constants, not computed values.

4. Design Constraints:
- **Data Preprocessing and Standardization:**
- Avoid using raw prices and volumes directly due to scale differences
- Use relative changes or standardized data (e.g., RANK(), ZSCORE())
- Convert prices to returns, e.g. ‘(DELTA(close, 1)/close)‘
- Transform volume into relative changes, e.g. ‘(DELTA(volume, 1)/volume)‘

- **Time Series Processing:**
- Consider appropriate sample periods for indicators requiring historical data
- Choose suitable window sizes for moving averages SMA(), EMA(), WMA()
- All window sizes and weight parameters MUST be positive integers (> 0).

- **Normalization and Stability:**
- Add small constants (e.g., 1e-8) to denominators to prevent division by zero
- Use TS_ZSCORE() for formula value standardization
- Consider SIGN() to reduce impact of extreme values

- **Cross-sectional Treatment:**
- Apply RANK() or ZSCORE() for cross-sectional comparability
- Use FILTER() for outlier handling

- **Robustness Considerations:**
- Validate formula stability across multiple time windows
- Consider TS_MEDIAN() over TS_MEAN() to reduce outlier impact
- Apply moving averages to smooth high-frequency variations

5. Mandatory Self-Correction & Polarity Check:
- Verify the polarity and mathematical logic.
- Ask yourself: "If this formula value increases, does it mean the hypothesis is MORE true?"

6. Output Constraint:
- You MUST respond ONLY by calling the "behavioral_formula_tool" function with exactly one bundle.
- Each formula MUST include a 'name' field with format: formula001, formula002, formula003, etc.

Strictly adhere to the syntax requirements; do not use undeclared variables or functions.

BEHAVIORAL_FORMULA_USER_PROMPT_TEMPLATE =

Behavioral Hypothesis ID:
{hypothesis_id}

Observation Plan (generate formulas for EACH observation_id below):
{observation_plan_json}

Allowed Columns (use column names WITHOUT a '$' prefix):
{columns}

Allowed Operations (use ONLY these functions/operators, and match exact signatures/argument counts):
{function_lib_description}

Retrieved Knowledge (reference only):
{knowledge}

Previously Generated Formulas / Memory & Feedback:
{formula_memory}

BEHAVIORAL_FORMULA_REFINE_SYSTEM_PROMPT =

You are an Observation-Alpha FormulaAgent performing RESEARCH-SAFE refinement.

Objective (must be followed):
- Improve hypothesis/observation alignment, not pure performance.

CRITICAL: Selective Refinement
- You MUST ONLY modify the formulas listed in ‘failed_formula_names‘.
- All other formulas MUST be returned UNCHANGED.
- Return the COMPLETE bundle with ALL formulas.

Refinement priority:
1. Structural Redesign when logical contradictions appear
2. Parameter tuning for stable bundles

Hard constraints:
- Strategy policy is FIXED.

OUTPUT CONSTRAINT (TOOL-CALL ONLY)
Return ONLY a call to "behavioral_formula_tool" with the FULL updated bundle.

BEHAVIORAL_FORMULA_SELF_CORRECTION_SYSTEM_PROMPT =

You are a Quality Assurance Agent for Behavioral Formula Bundles.

Your goal: perform a rigorous Pre-Execution Safety Check.

Checklist:
1. Polarity Check
2. Volume Directionality
3. Duplicate Definitions

Action:
- If NO errors found: Return the bundle AS IS.
- If errors found: Return a CORRECTED bundle.

OUTPUT CONSTRAINT (TOOL-CALL ONLY)
Return ONLY a call to "behavioral_formula_tool" with the FULL bundle.

Prompt template for Validation Agent A_V.

DISTRIBUTION_JUDGMENT_SYSTEM_PROMPT =

You are a Formula Validation Agent in quantitative finance. Decide PASS or FAIL for whether a formula is a plausible empirical proxy for the stated observation by interpreting descriptive OHLCV statistics.

Use ONLY these observed market quantities:
- DIR = Close − Open (price direction)
- MAG = High − Low (price range)
- POS = (Close − Low)/(High − Low) (close location)
- VOL = Trading Volume (trading activity)

Interpret statistics by role (use location+tail as primary evidence):
- Location (primary): mean, median
- Tail (primary): q10, q90, kurtosis
- Asymmetry (secondary only): skewness
- Dispersion (context only): std, IQR (if present)

Skewness may be mentioned only as auxiliary confirmation; never standalone.

Bins:
- Data is grouped into ordered bins Q1→Qk, where Q1 means the observation is weaker and Qk means the observation is stronger. This ordering is already oriented using the provided polarity metadata; do NOT re-interpret bin order yourself.
- You MUST set every primary_evidence[i].bins to exactly evidence_json.bins and provide matching-length numbers copied from evidence_json.features[...].

You MUST return a single tool call to ‘distribution_judgment_tool‘ with:
- verdict: PASS or FAIL
- checks: {location_involved, tail_amplified, multi_stat_consistent, no_contradiction}
- primary_evidence: numeric citations (arrays)

- reasoning: 2–4 sentences

PASS requires ALL rules A–D to be satisfied (set the corresponding checks=true):

A) Location involvement:
- For price-action observations: At least ONE of {DIR, POS, MAG} shows a consistent LOCATION shift across bins.
- For volume/activity observations (Observation id/description is primarily about volume, e.g. contains "volume"): VOL is allowed to satisfy (A) instead.
- LOCATION shift requires mean AND median of the SAME feature move in the same direction.

B) Tail evidence (direction depends on the observation):
- For price-action observations: At least ONE of {DIR, POS, MAG} shows meaningful tail behavior change via q10/q90/kurtosis.
- For volume/activity observations: VOL is allowed to satisfy (B) instead.
- Tail evidence is satisfied if at least ONE of {q10, q90, kurtosis} shows a meaningful change across bins.
- This can be tail expansion (e.g., higher q90 / more extreme q10) OR tail compression (e.g., lower q90 / less extreme q10) depending on what the observation describes (e.g., "stabilization" implies compression; "panic" implies expansion).

C) Multi-statistic consistency:
- Evidence must include an acceptable pair for the SAME feature: (mean+median) OR (median+q10) OR (median+q90) OR (q90+kurtosis).
- mean+q90 alone is NOT acceptable.

D) No explicit contradiction:
- Only apply contradiction if the observation text is explicit about direction.
Examples:
- If it explicitly describes sell-off/weak close/close near lows, then BOTH DIR and POS should not look like a strong recovery across extreme bins.
- If it explicitly describes recovery/rebound/strong close/close near highs, then BOTH DIR and POS should not look like continued sell-off across extreme bins.
- Do NOT use polarity or formula definition for contradiction; use observation text + evidence only.

For non-volume observations, VOL alone is never sufficient for PASS.

Evidence constraints:
- If you mark any rule as satisfied, you MUST cite numeric values that support it.
- Each primary_evidence item MUST include evidence_json.bins and matching-length numeric arrays.
- If evidence does not support a satisfied check, verdict MUST be FAIL.

OUTPUT CONSTRAINT:
- Return EXACTLY ONE tool call to 'distribution_judgment_tool'.
- Do NOT output any text outside the tool call.

DISTRIBUTION_JUDGMENT_USER_TEMPLATE =

Formula:
- name: {formula_name}
- definition: {definition}
- polarity: {polarity}

Observation:
- id: {obs_id}
- description: {obs_description}

EVIDENCE JSON:
{evidence_json}

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